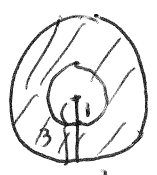
Math 241, F14 **Name:**

Quiz 12 **SSN: xxx-xx-**

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.**

Problem 1. (5 points) Using the cylindrical coordinates to compute the triple integral $\iiint_E z\sqrt{x^2+y^2}\,dV$ where E is the solid that lies between the cylinders $x^2+y^2=1$ and $x^2+y^2=9$, and between the planes $z=-2$ and $z=4$.

$$\iiint_E z\sqrt{x^2+y^2}\,dV = \int_0^{2\pi}\int_1^3\int_{-2}^4 r z\cdot r\,dz\,dr\,d\theta$$



$$= \int_0^{2\pi}\int_1^3 r^2\left[\frac{z^2}{2}\right]_{z=-2}^{z=4} dr\,d\theta$$

$E=\{(r,\theta,z)\mid 0\le\theta\le 2\pi,\ 1\le r\le 3,\ -2\le z\le 4\}$

$$= \int_0^{2\pi}\int_1^3 6r^2\,dr\,d\theta$$

$$= \int_0^{2\pi}\left[2r^3\right]_{r=1}^{r=3} d\theta$$

$$= \int_0^{2\pi} 52\,d\theta = 104\pi$$

Problem 2. (5 points) Evaluate $\iiint_E \sqrt{x^2+y^2+z^2}\,dV$ where $E=\{(\rho,\theta,\phi)\mid 0\le\rho\le 2,\ 0\le\theta\le\pi/2,\ \pi/3\le\phi\le\pi/2\}$ using the triple integral in spherical coordinates.

$$\iiint_E \sqrt{x^2+y^2+z^2}\,dV = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\int_0^{\pi/2}\int_0^2 \rho\cdot\rho^2\sin\phi\,d\rho\,d\theta\,d\phi$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}}\int_0^{\pi/2}\sin\phi\left[\frac{\rho^4}{4}\right]_{\rho=0}^{\rho=2} d\theta\,d\phi$$

$$= \int_{\pi/3}^{\pi/2}\int_0^{\pi/2} 4\sin\phi\,d\theta\,d\phi$$

$$= \int_{\pi/3}^{\pi/2} 2\pi\sin\phi\,d\phi = 2\pi\left[-\cos\phi\right]_{\phi=\pi/3}^{\phi=\pi/2}$$

$$= \pi$$